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Error-Corrected Inference-Time Scaling for Imperfect Diffusion Models

The Energy-based Feynman-Kac Corrector Removes Systematic Model Bias at Inference Time, Achieving Accurate Molecular Free-Energy Profiles Where Standard Particle Methods Fail

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Inference-time scaling for diffusion models—running more particles through sequential Monte Carlo—reduces variance but cannot remove the systematic bias introduced by an imperfect pretrained score function. The Energy-based Feynman-Kac Corrector (EBFKC) derives dynamics that track a prescribed probability path exactly in the population limit regardless of model error, and demonstrates accurate molecular free-energy profiling on alanine dipeptide and tetrapeptide where standard methods fail.

AT A GLANCE

Problem: Standard inference-time scaling inherits the systematic bias of an imperfect diffusion model; more particles reduce variance but not bias.

Why it matters: Scientific applications (molecule generation, free-energy computation) require accurate tail distributions, not just good mode coverage.

Method: Feynman-Kac potential derived from an energy-based reference corrects the imperfect score on the fly; approximated via sequential Monte Carlo with variance-controlling guidance.

Result: Closely matches ground-truth free-energy surfaces on alanine dipeptide and tetrapeptide; standard SMC baselines retain several $k_B T$ of error.

Evidence: Gaussian mixture models, particle systems, two alanine peptides under annealing and reward tilting.

The Big Picture

Inference-time scaling has emerged as a key technique for improving diffusion model quality: by running a sequential Monte Carlo (SMC) procedure with many particles, one can steer generation toward a target distribution or reward signal without retraining the model. Applications range from image generation with aesthetic rewards to molecular

conformation sampling with energy-based targets.

All existing approaches, however, share a silent assumption: that the pretrained diffusion model’s score function is exact. In practice, score functions are trained on finite data with finite compute and carry systematic approximation error. Standard SMC reduces the statistical error of the particle approximation but cannot remove this model-level bias: every particle evolves under the same imperfect dynamics, so the distribution they jointly represent converges to the wrong target.

Why Existing Approaches Fall Short

Monte Carlo only reduces variance, not bias. Adding more particles in standard SMC reduces $O(1/\sqrt{N})$ variance, leaving a fixed systematic error proportional to the score function mismatch $\epsilon_t = \hat{s}_\theta - \nabla \log p_t$.

Path tracking error. Imperfect scores cause particles to drift off the prescribed probability path, accumulating errors that compound over diffusion timesteps.

Reward tilting. Reward-guided diffusion reweights samples assuming a correct model; systematic score error introduces additional bias into the tilted distribution that no reweighting scheme can correct post-hoc.

Core Mechanism

Energy-based Feynman-Kac Corrector (EBFKC). Given a reference energy $E_t(\mathbf{x})$ defining the desired path $\pi_t \propto e^{-E_t(\mathbf{x})}$, the paper derives a Feynman-Kac potential G_t that, when incorporated as a multiplicative weight in the SMC procedure, ensures the particle ensemble tracks the prescribed path $\{p_t\}$ exactly in the continuous-time population limit, regardless of the score function error ϵ_t .

The key insight is that the Feynman-Kac formula from stochastic analysis provides a closed-form expression for the correction term that neutralizes the effect of ϵ_t without requiring access to the true score function—only the reference energy is needed.

Technical Formulation

Let $\hat{s}_\theta(\mathbf{x}, t)$ be the imperfect score with error $\epsilon_t(\mathbf{x}) = \hat{s}_\theta - \nabla \log p_t$. Standard SMC propagates under:

$$d\mathbf{x}_t = \hat{s}_\theta(\mathbf{x}_t, t) dt + \sqrt{2} dW_t$$

and tracks p_t only when $\epsilon_t = 0$.

EBFKC derives the Feynman-Kac potential:

$$G_t(\mathbf{x}) = \exp\left(-\int_0^t \langle \epsilon_\tau(\mathbf{x}_\tau), \nabla \log p_\tau(\mathbf{x}_\tau) \rangle d\tau\right)$$

The corrected particle system with weights proportional to G_t converges to p_t in the population limit. A variance-

controlling guidance term:

$$\hat{s}_\theta^{\text{VC}}(\mathbf{x}, t) = \hat{s}_\theta(\mathbf{x}, t) - \gamma \nabla_{\mathbf{x}} \log w_t(\mathbf{x})$$

prevents particle collapse by penalizing low-weight regions, where $\gamma > 0$ is a guidance strength parameter.

Learning or Inference Procedure

EBFKC requires no additional training—it operates on a frozen pretrained diffusion model.

1. **Specify** reference energy E_t (free energy functional or reward model).
2. **Initialize** N particles $\{\mathbf{x}_T^{(i)}\}_{i=1}^N$ from the prior p_T .
3. **Propagate** with $\hat{s}_\theta$ plus variance-controlling guidance, accumulating Feynman-Kac weights $G_t^{(i)}$ at each step.
4. **Resample** when the effective sample size $N_{\text{eff}} = (\sum_i w_i^2)^{-1}$ drops below $N/2$.
5. **Return** the weighted ensemble $\{(\mathbf{x}_0^{(i)}, w_0^{(i)})\}$.

The overhead relative to standard SMC is one energy evaluation per particle per step.

What the Guarantee Says

In the continuous-time population limit ($N \rightarrow \infty$, $\Delta t \rightarrow 0$), EBFKC tracks the prescribed path $\{p_t\}$ exactly, independent of ϵ_t . For finite N , the error scales as $O(1/\sqrt{N})$ in particle count and $O(\Delta t)$ in discretization step size. Crucially, unlike standard SMC, the $N \rightarrow \infty$ limit converges to the correct target rather than a biased surrogate.

Experimental Findings

- **Gaussian mixture models:** EBFKC recovers all modes of a multi-modal 2D target; standard SMC with imperfect model overweights dominant modes.
- **Particle systems:** EBFKC accurately recovers equilibrium distributions for interacting particles; standard baselines show persistent density errors.
- **Alanine dipeptide:** EBFKC matches the ground-truth Ramachandran free-energy surface within statistical uncertainty; standard SMC exhibits errors of several $k_B T$ in high-barrier regions.
- **Alanine tetrapeptide:** EBFKC scales to the larger system and maintains accurate free-energy profiles under both annealing and reward-tilting regimes.

Ablations and Interpretation

Particle count: EBFKC converges faster to the correct target than standard SMC as N increases; at low N , variance-controlling guidance is essential.

Model imperfection level: As score error ϵ_t is arti-

ficially increased, standard SMC degrades rapidly while EBFKC remains accurate.

Variance-controlling guidance: Removing this term causes particle collapse at long horizons, confirming its necessity.

Feynman-Kac vs. naive reweighting: Simple importance weighting without the Feynman-Kac path correction tracks a biased distribution even with large N , underscoring that the correction must operate on the dynamics, not just the terminal weights.

Reference

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